Lesson 4: V=IR Graphs

Current-Potential Difference Characteristics of Components

Teacher: Dan (1 Practical)

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Starter: Power Up Your Memory

What does the gradient of a current-voltage (I-V) graph represent?

Success Cue: Gradient = ΔI / ΔV

Answer: 1 / Resistance (steeper gradient = lower resistance)

What shape is the I-V graph for a fixed resistor at constant temperature?

Success Cue: Directly proportional

Answer: Straight line through the origin

Why does the resistance of a filament lamp increase as current increases?

Success Cue: Temperature rise

Answer: Filament heats up, ions vibrate more, more electron collisions

In which direction does a diode allow electric current to flow?

Success Cue: Arrow in symbol

Answer: Forward direction only

What is an 'ohmic conductor'?

Success Cue: Obeys Ohm's law

Answer: A conductor where current is directly proportional to voltage at constant temp

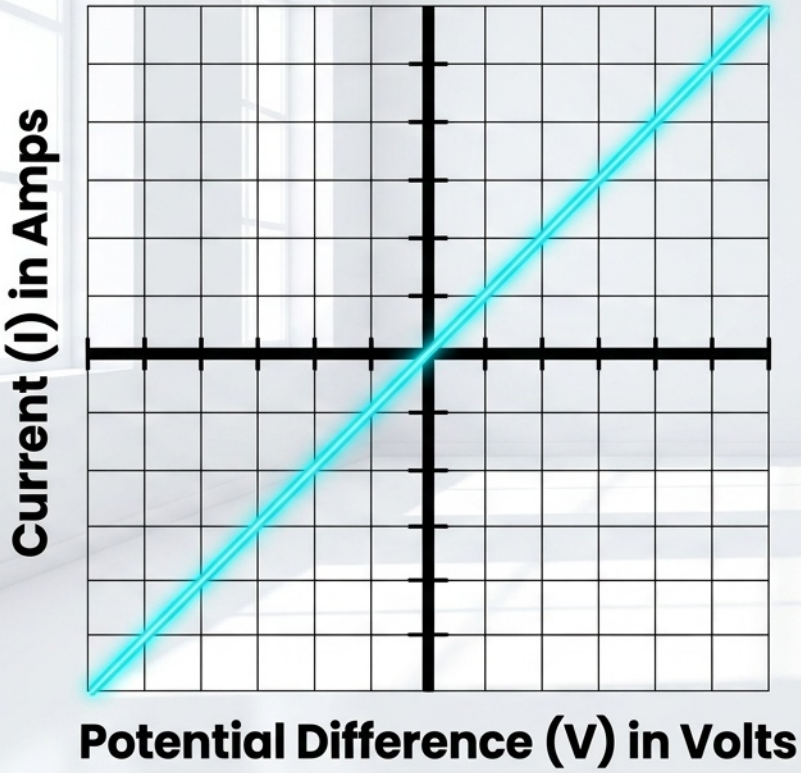
What does the reverse-bias region of a diode I-V graph look like?

Success Cue: Blocks reverse current

Answer: Flat zero line along the voltage axis (very high resistance)

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What is an I-V Graph?



The Axes

Current (I) is on the vertical y-axis.
Voltage (V) is on the horizontal x-axis.

The Gradient

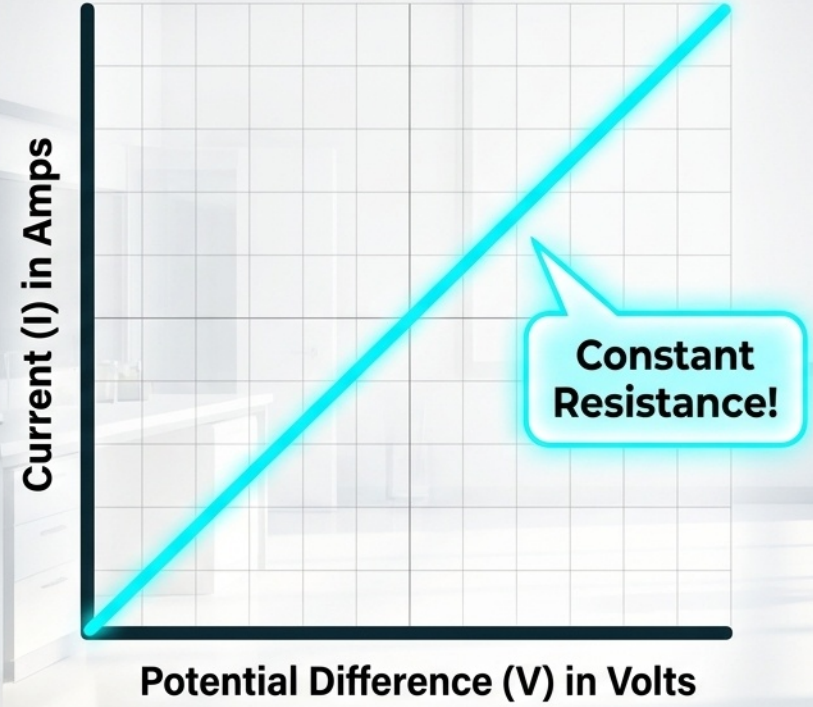
Steeper line = Higher current for the same voltage.
Therefore, **Steeper line = Lower resistance**.

The Origin

Passing through (0,0) means zero voltage equals zero current.

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1. Ohmic Conductor (Fixed Resistor)



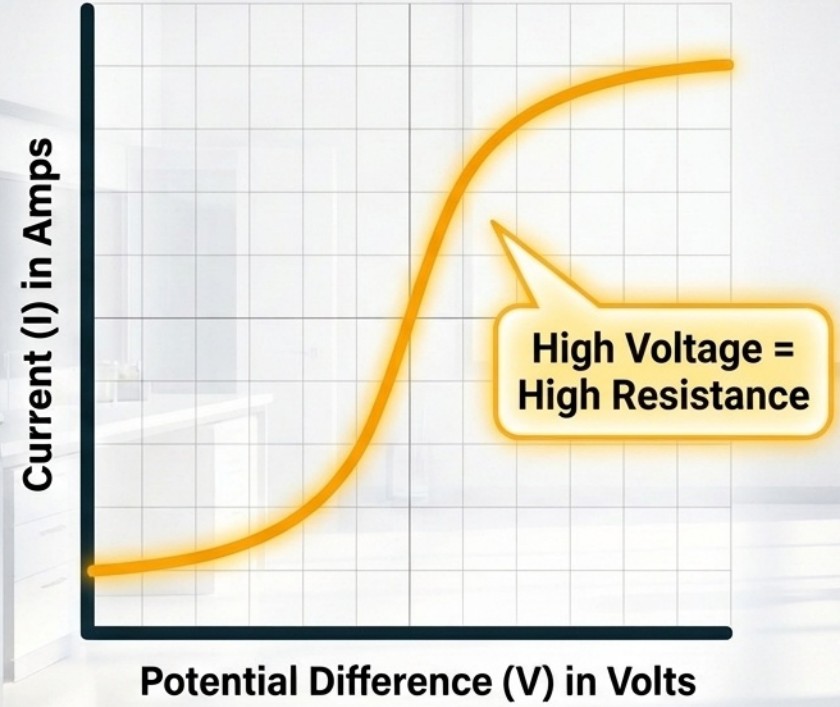


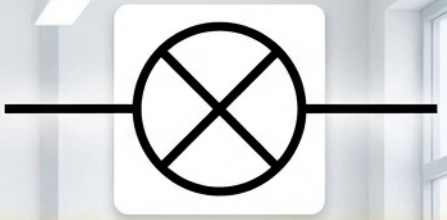
The Shape:
- Straight line directly through the origin.
- Constant gradient across all voltages.

The Physics:
- Current is directly proportional to voltage.
- Resistance remains constant (at a constant temperature).
- Obeys Ohm's Law perfectly.

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2. Filament Lamp





The Shape:
- S-shaped curve passing through the origin.
- Gradient flattens out at higher voltages.

The Physics:
- Current does not increase proportionally.
- As voltage increases, resistance increases.
- The flat curve means it is harder for current to flow.

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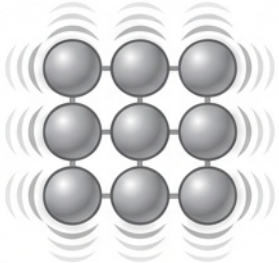
Inside the Filament: Why Resistance Rises

Heating Up



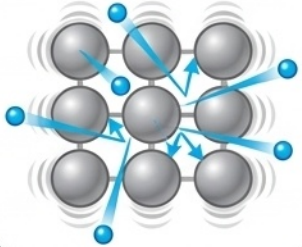
High current causes filament temperature to rise rapidly.

Vigorous Vibrations



Heat causes metal lattice ions to vibrate more vigorously.

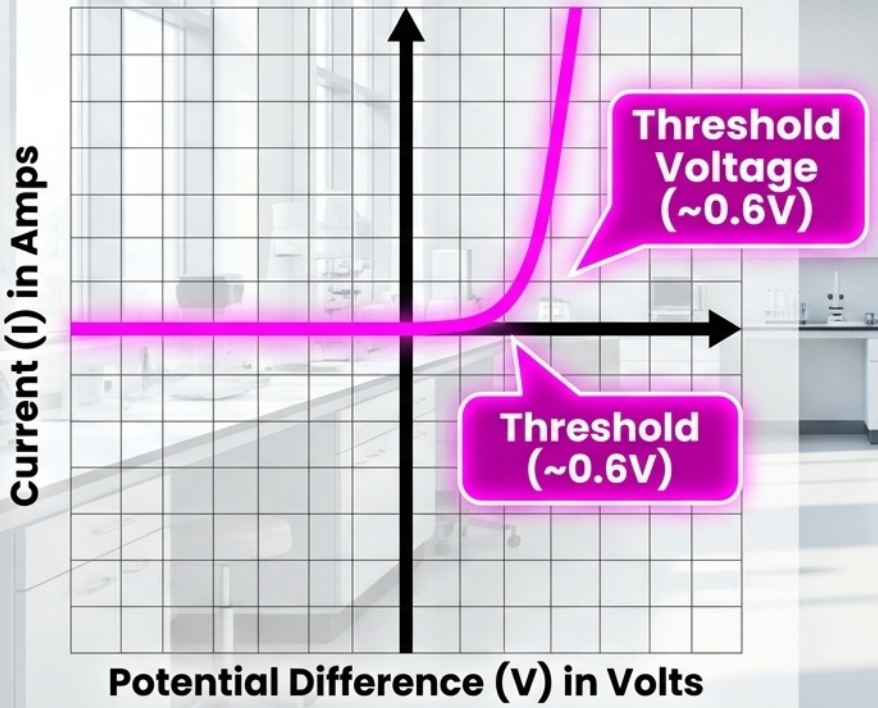
Electron Collisions



Electrons collide more frequently. Resistance increases, flattening the graph.

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3. Semiconductor Diode & LED



The Shape:
- Flat zero line, then a sharp upward curve.

Forward Bias (Right):
- Current flows above threshold voltage (~0.6 V).
- Resistance drops rapidly.

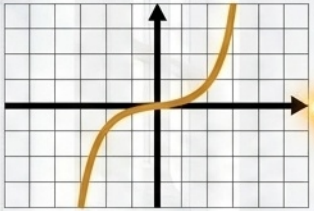
Reverse Bias (Left):
- Current is completely blocked.
- Extremely high resistance (negligible current).

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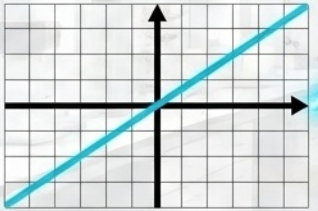
Checkpoint: Match the Signature

Pair-and-Share. Match the component to its unique I-V graph shape.
You have 60 seconds!

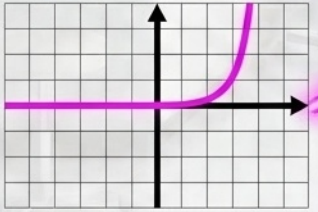
A



B



C



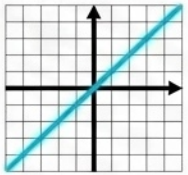
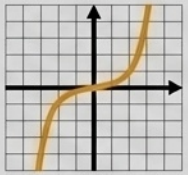
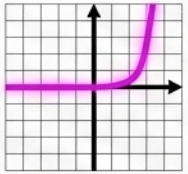
Fixed Resistor

Semiconductor Diode

Filament Lamp

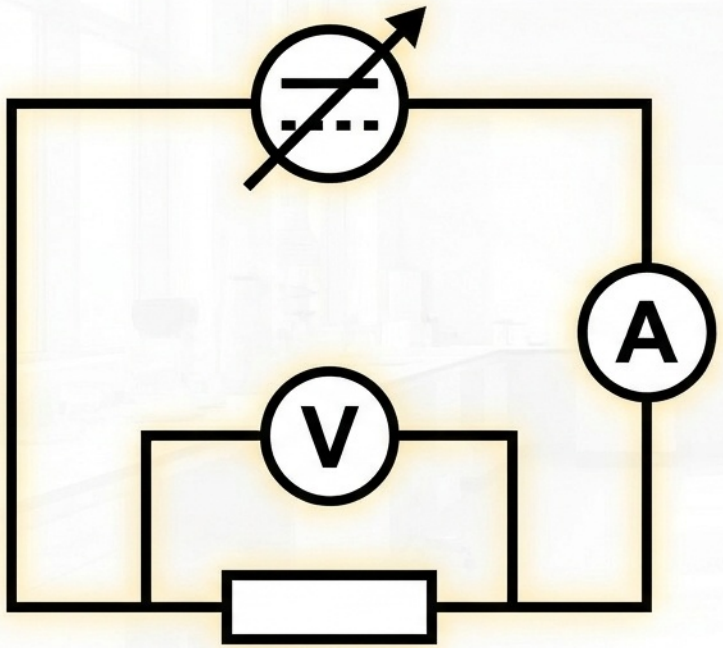
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Synthesis: The I-V Characteristics

Component	Graph Shape	Resistance Behaviour
Fixed Resistor 	Straight line through origin	Constant (at constant temp)
Filament Lamp 	S-shaped curve	Increases as current increases
Diode / LED 	Flat zero, then sharp rise	Very high in reverse, Very low forward

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Practical 1: I-V Characteristics



The Objective:

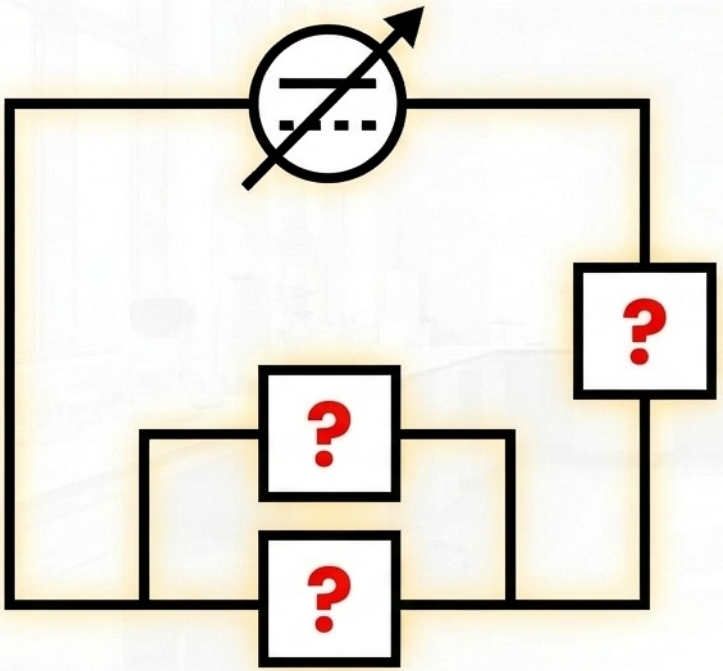
- Plot I-V graphs for a resistor, lamp, and diode.

The Equipment:

- Variable DC Supply (to change V)
- Ammeter (in series, to measure I)
- Voltmeter (in parallel, to measure V)
- Test Components (Resistor, Lamp, Diode)
- Connecting Leads

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Checkpoint: Label the Setup



Work with your lab partner. Identify the missing components.

- Where does the Voltmeter go?
- Where does the Ammeter go?
- Why?

Ammeter in series, Voltmeter in parallel across the component.

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Practical Workflow: Step-by-Step

Step 1: Build the Circuit

Connect supply, ammeter, and component in series. Add voltmeter in parallel.

Step 2: Start Low

Set the variable power supply to 0 V. Record Current (I) and Voltage (V).

Step 3: Increase and Record

Increase voltage in small increments (e.g., 1 V). Record pairs of V and I.

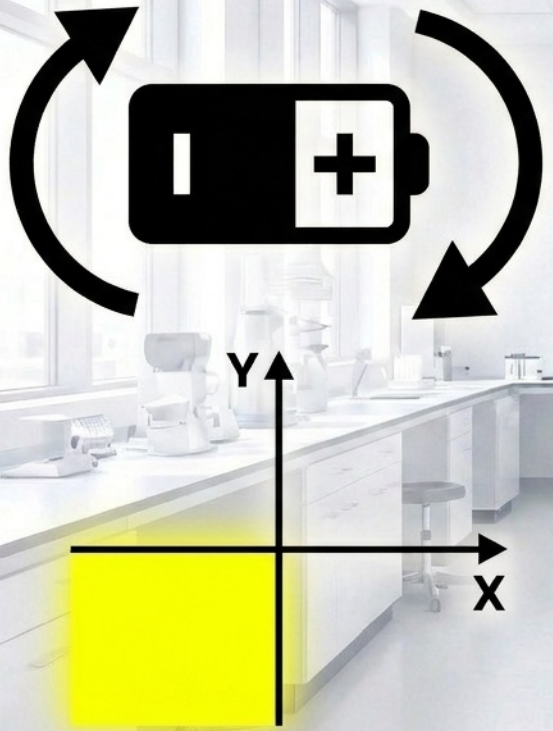
Step 4: Swap Components

Repeat the entire process for the Lamp and the Diode.

SAFETY! Turn off supply between readings to prevent overheating the resistor!

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Practical: Reversing the Polarity



Why negative values?

- To see component behaviour when current flows backwards.

How do we do it?

- Reverse the connections at the DC power supply.

What happens to the meters?

- Voltmeters and ammeters will now read negative values.
- Plot these to build the left side of your graph.

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Challenge: Calculate the Resistance!

Formula Reminder: R = V / I

Scenario A (Resistor)

V = 4.0 V

I = 0.5 A

Resistance = ?

8 Ω

Scenario B (Filament Lamp)

V = 8.0 V

I = 0.4 A

Resistance = ?

20 Ω

First pair to calculate both correctly wins!

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Summary & Final Check

Ohmic Conductor

- Straight line through origin.
- Constant resistance.

Filament Lamp

- S-shaped curve.
- Resistance rises with heat.

Semiconductor Diode

- Current in one direction only.
- High resistance in reverse.

Exit Ticket: Turn to your partner and explain *why* the filament lamp curve flattens out at high voltages.

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1. Fixed Resistor Ohmic | 2. Filament Lamp Non-Ohmic | 3. Semiconductor Diode Non-Ohmic

EXPERIMENTAL RESULTS TABLE

7 / 7 PLOTTED

Status	P.D. (V)	Current (A)	R = V/I (Ω)
●	-6.0	-0.600	10.0 Ω
●	-4.0	-0.400	10.0 Ω
●	-2.0	-0.200	10.0 Ω
●	0.0	0.000	10.0 Ω
●	2.0	0.200	10.0 Ω
●	4.0	0.400	10.0 Ω
●	6.0	0.600	10.0 Ω

GRAPH PLOTTING ACTIONS

Plot Next Point (+) | Plot All Points

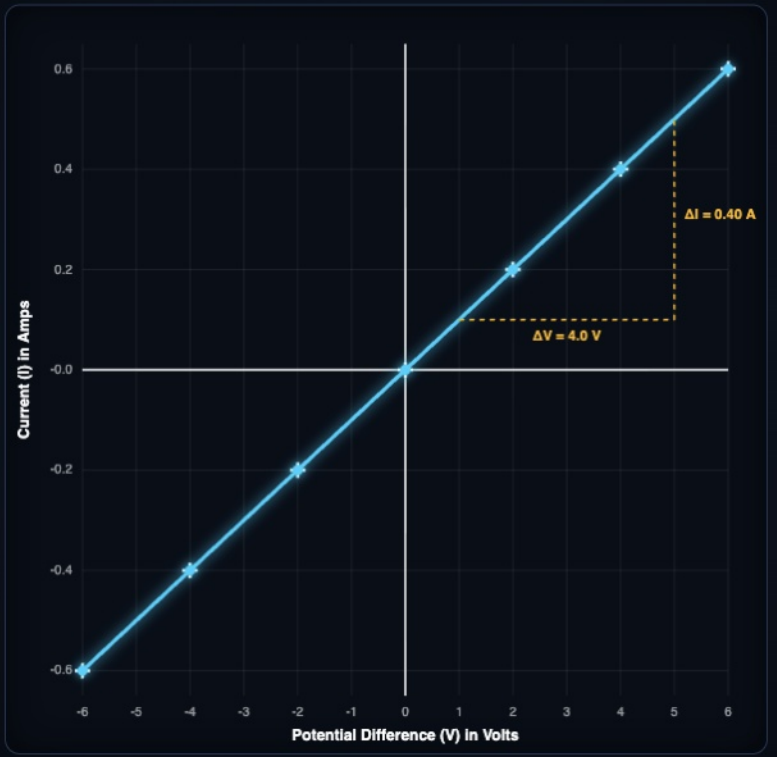
Clear Plotted Points

ANALYSIS OVERLAYS

- Line of Best Fit
- Gradient & Resistance Triangle
- Show Negative Quadrant (Reversed Polarity)

Fixed Resistor (10 Ω) OHMIC CONDUCTOR

Hover points to inspect coordinates · Scale: 2mm grid calibration



CALCULATED LIVE METRICS

0.100 A/V
GRADIENT (ΔI / ΔV)

10.0 Ω
EFFECTIVE RESISTANCE (1/M)

THE PHYSICS INSIDE THE COMPONENT

Fixed Resistor: Current is directly proportional to potential difference. The gradient remains completely constant because the temperature is kept constant. It obeys Ohm's Law (V = IR) across all voltages.

Checkpoint Challenge

Target: 100%

What does the constant straight-line gradient through the origin tell us about this resistor?

- Current is directly proportional to voltage; resistance is constant.
- Resistance increases as the voltage increases.
- Current flows only in the forward direction.

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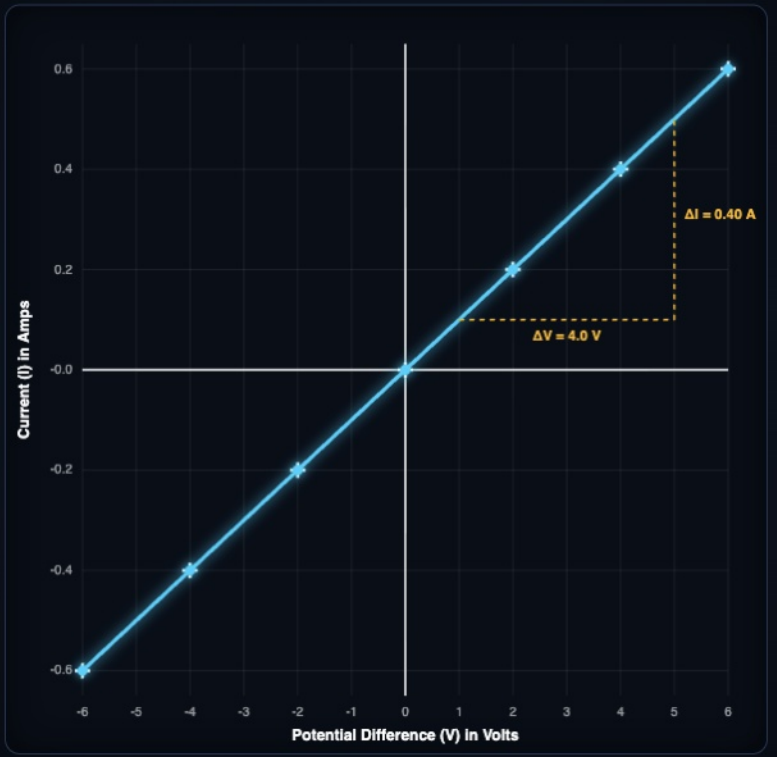
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